

OSCODA-WURTSMITH, MI, USA

WMO: **726395**Lat: **44.450N**Lon: **83.400W**Elev: **193**StdP: **99.03**Time Zone: **-5.00 (NAE)**Period: **97-19**WBAN: **14808**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-17.2	-23.8	0.4	-19.6	-21.3	0.6	-15.6	11.6	-4.4	10.7	-4.4	3.1	240	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.0	22.2	29.9	21.3	27.9	20.3	23.9	29.3	22.8	27.7	21.8	26.0	5.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.4	26.8	21.3	16.3	25.5	20.4	15.4	24.4	72.4	29.7	68.2	27.9	64.4	26.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	8.2	-24.0	34.6	3.7	2.0	-26.6	36.0	-28.8	37.2	-30.8	38.3	-33.5	39.7		
(5)			-24.3	25.5	3.6	1.3	-26.9	26.4	-29.0	27.2	-31.0	27.9	-33.6	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-6.0	-5.3	-0.8	5.4	12.0	17.5	20.6	19.7	16.0	9.3	3.2	-2.6
	DBStd	10.55	6.02	5.79	5.84	4.88	4.89	4.26	3.60	3.21	4.29	4.70	4.66	5.29
	HDD10.0	2126	497	429	337	155	32	1	0	0	3	72	209	391
	HDD18.3	4192	755	663	592	388	207	67	18	21	96	282	454	649
	CDD10.0	1207	0	0	3	17	93	226	327	302	183	51	4	1
	CDD18.3	232	0	0	0	1	10	42	87	64	26	3	0	0
	CDH23.3	2201	0	0	2	18	144	448	819	531	214	24	0	0
	CDH26.7	697	0	0	1	3	44	158	278	151	57	6	0	0
Wind	WSAvg	3.6	4.1	4.1	3.9	4.1	3.6	3.1	3.0	3.0	3.2	3.8	3.9	4.0
Precipitation	PrecAvg	768	45	38	43	68	78	85	84	78	72	70	60	45
	PrecMax	937	76	81	124	120	168	177	162	151	129	162	134	102
	PrecMin	629	6	10	12	29	34	23	28	31	11	21	18	11
	PrecStd	80	22	19	27	22	34	36	31	29	31	35	34	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	12.0	20.9	26.2	31.1	32.9	34.1	32.8	31.3	26.3	18.5	12.9
		MCWB	7.7	10.1	12.4	15.4	19.9	22.6	23.6	22.6	22.1	18.1	13.3	11.0
	2%	DB	6.1	7.0	14.1	21.1	27.5	31.1	32.1	30.3	27.9	22.4	15.1	8.9
		MCWB	4.2	4.3	8.8	13.2	17.8	21.5	22.6	22.3	20.5	16.7	11.2	7.1
	5%	DB	3.6	4.0	11.0	17.3	24.0	28.0	29.8	27.9	26.0	19.6	12.5	6.3
		MCWB	2.3	2.1	7.6	11.3	16.5	20.1	21.3	20.8	19.4	15.8	9.7	4.6
	10%	DB	2.3	2.5	7.4	13.7	21.2	26.0	27.7	26.3	23.5	17.4	10.4	4.1
		MCWB	1.3	1.1	4.3	8.6	15.0	18.8	20.5	19.9	18.4	13.9	8.1	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	9.8	15.0	17.4	21.6	24.1	25.7	24.7	23.6	21.0	14.6	11.9
		MCDB	8.3	12.5	19.0	23.2	27.7	31.0	31.9	29.5	28.9	24.3	17.3	13.0
	2%	WB	4.5	4.5	10.6	14.5	19.7	22.8	24.3	23.4	22.1	18.1	12.0	7.8
		MCDB	5.9	6.6	14.0	19.3	24.4	28.8	29.6	28.5	26.1	20.5	14.4	8.6
	5%	WB	2.6	2.6	7.4	11.7	17.9	21.4	22.9	22.2	20.9	16.2	10.0	5.0
		MCDB	3.5	4.0	9.9	16.4	22.5	26.4	27.4	26.3	23.7	19.2	12.1	6.3
	10%	WB	1.2	1.0	4.4	9.2	15.7	19.9	21.8	21.1	19.6	14.4	8.1	3.1
		MCDB	2.2	2.4	7.3	13.4	20.6	24.4	25.9	24.7	22.7	17.0	9.8	4.4
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.3	9.9	10.9	11.9	12.0	11.9	11.3	11.7	10.1	8.4	7.3
		MCDBR	7.7	9.6	14.5	17.0	16.2	14.9	13.8	13.4	14.0	13.8	11.7	8.9
		MCWBR	6.9	7.7	10.1	10.4	8.8	7.4	6.4	6.8	7.5	8.9	8.9	7.6
	5% WB	MCDBR	7.2	8.8	13.4	15.3	14.3	13.3	11.8	11.6	11.6	12.0	10.7	8.3
		MCWBR	6.7	7.6	10.0	10.5	8.8	7.7	6.2	6.4	6.9	8.6	8.7	7.5
Clear-Sky Solar Irradiance	taub		0.281	0.299	0.305	0.351	0.374	0.409	0.407	0.400	0.372	0.344	0.311	0.281
	taud		2.451	2.373	2.463	2.380	2.387	2.317	2.342	2.355	2.425	2.483	2.524	2.500
	Ebn at Noon		850	891	937	912	896	862	858	852	851	831	809	817
	Edh at Noon		71	91	96	114	117	126	121	116	100	82	67	62
All-Sky Solar Radiation	RadAvg		1.31	2.10	3.61	4.58	5.45	6.00	6.05	5.14	4.07	2.46	1.43	1.01
	RadStd		0.10	0.28	0.38	0.39	0.48	0.38	0.25	0.21	0.38	0.27	0.19	0.10